

Empirical local synchrony range

Adaptive-range versus fixed 100 km synchrony

HOLD

The method is implemented and validated,
but the real-data scale-up gate fails.

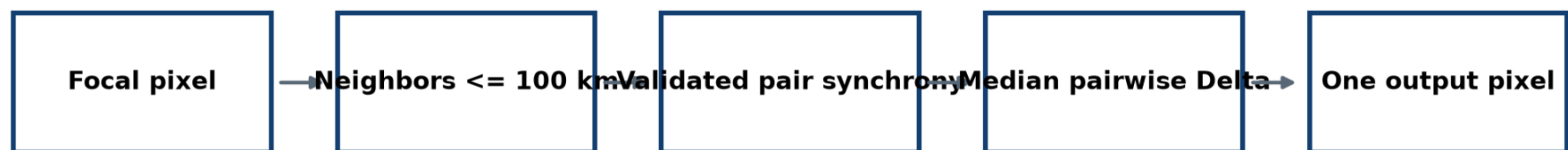
- 0 of 5 national representative pixels resolved a common range at 500 km / 20 km bins.
- 1 of 25 Colorado pilot sites resolved a common cold/warm range.
- 96% of Colorado common estimates were censored or unresolved.
- 0 of 25 stabilized between 400 and 500 km.

Existing fixed 100 km method

1. CONTROL

The validated control calculates physical-distance pairs once, applies the established cold/warm tail-Spearman kernel, excludes self-pairs, and immediately collapses pairwise Delta to one focal-pixel value. This result is preserved unchanged.

Existing fixed 100 km control



No stacking | no second convolution | self-pair excluded

Why test an adaptive range?

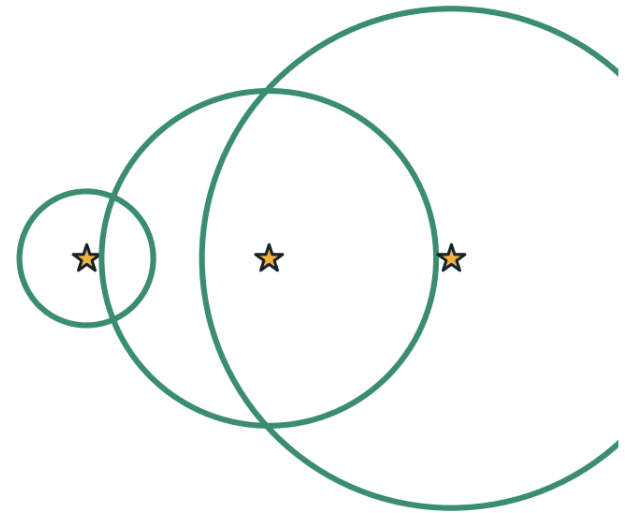
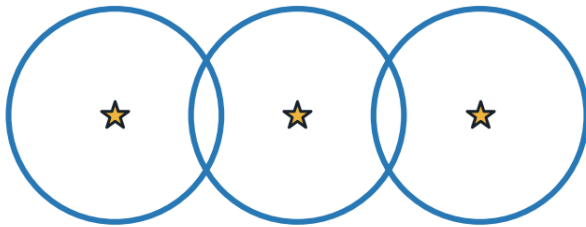
2. SCIENTIFIC QUESTION

A uniform 100 km radius is a defensible control, but it may cross distinct terrain and climate regimes in one place while truncating coherent structure elsewhere. The experiment changes only which neighbors enter the same robust collapse.

The experiment changes neighbor selection only

Fixed: every focal pixel gets 100 km

Question: does empirical support differ?

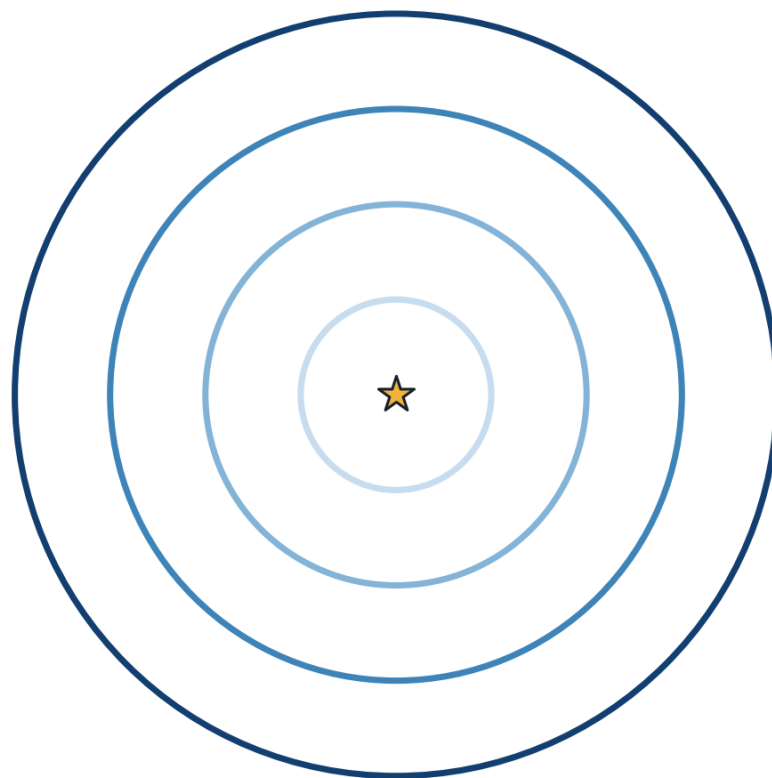


Start with one focal pixel

3. LOCAL SCIENTIFIC OBJECT

Every estimate is focal-centered. Distance is measured in great-circle kilometers from the focal cell to eligible PRISM cells. Raster-cell counts and angular degrees are never treated as range units.

One focal pixel inside large discovery support

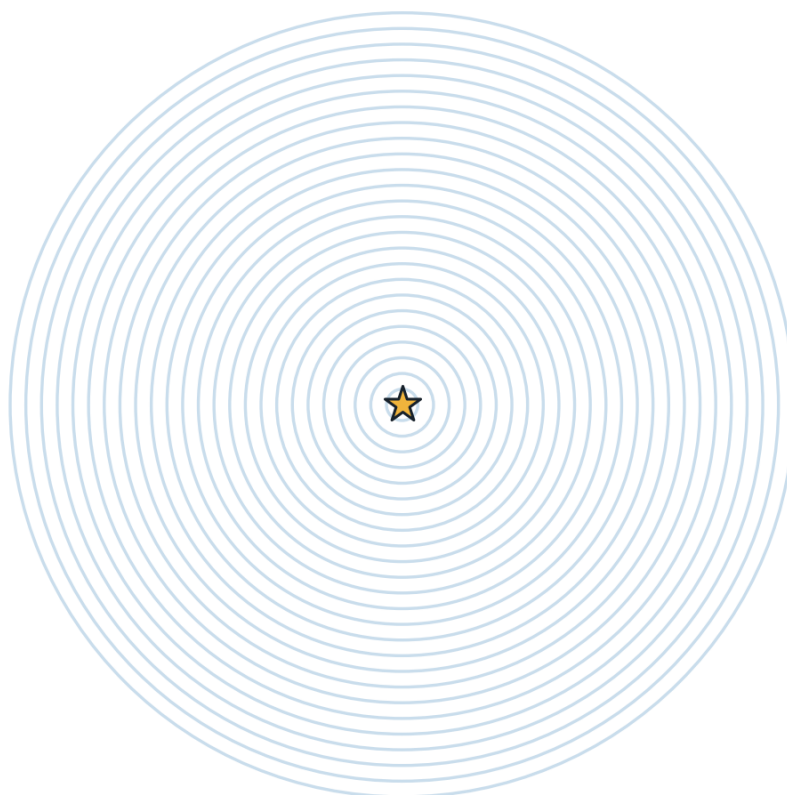


Observe well beyond the candidate range

4. DISCOVERY SUPPORT

The pilot uses a 500 km discovery radius so the empirical curve has an opportunity to approach a distant background. Discovery support is an observation limit, not an inferred range; reaching it produces a censoring flag.

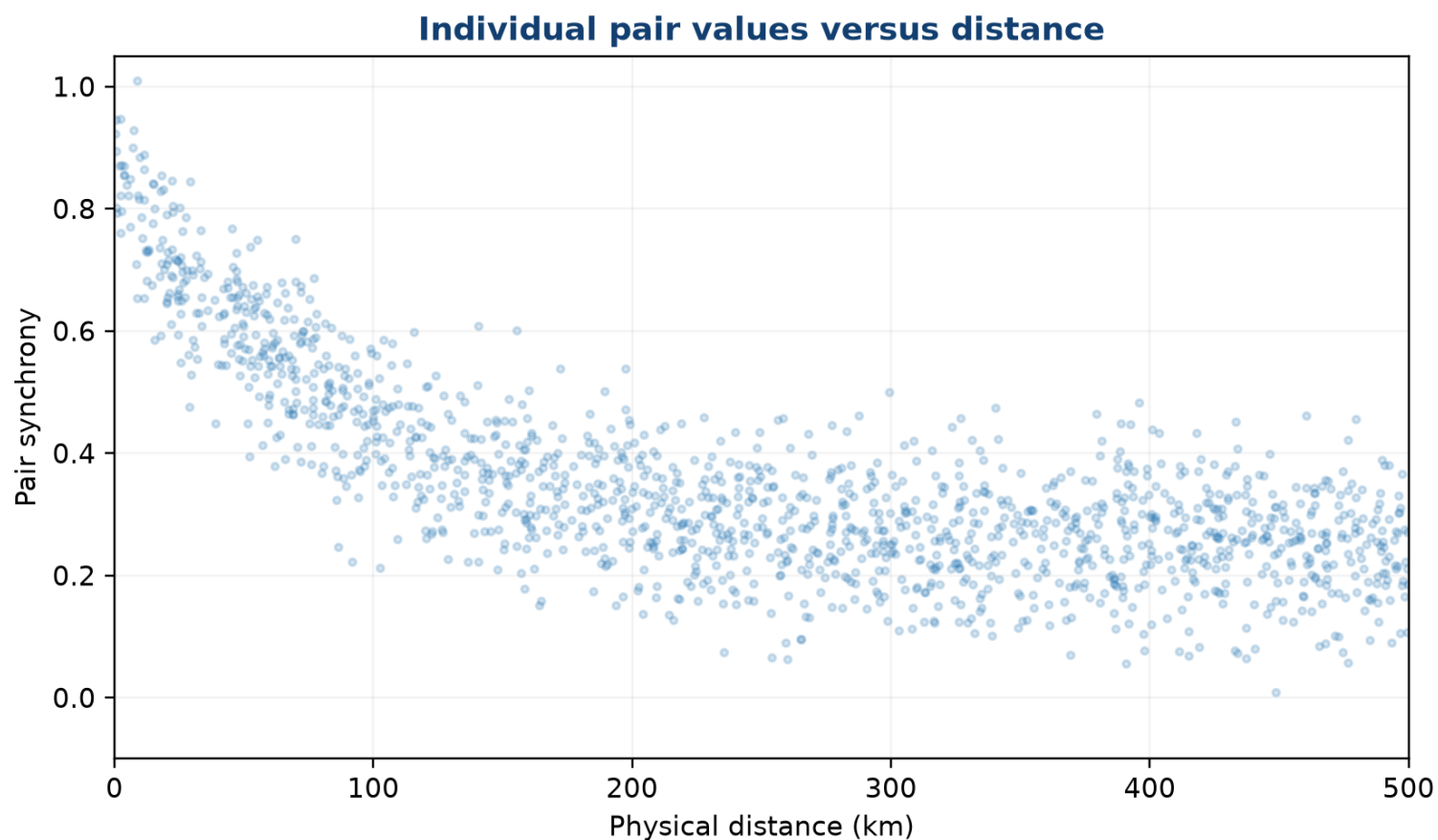
Twenty-kilometer physical annuli



Reuse every pair relationship

5. PAIR PRODUCT

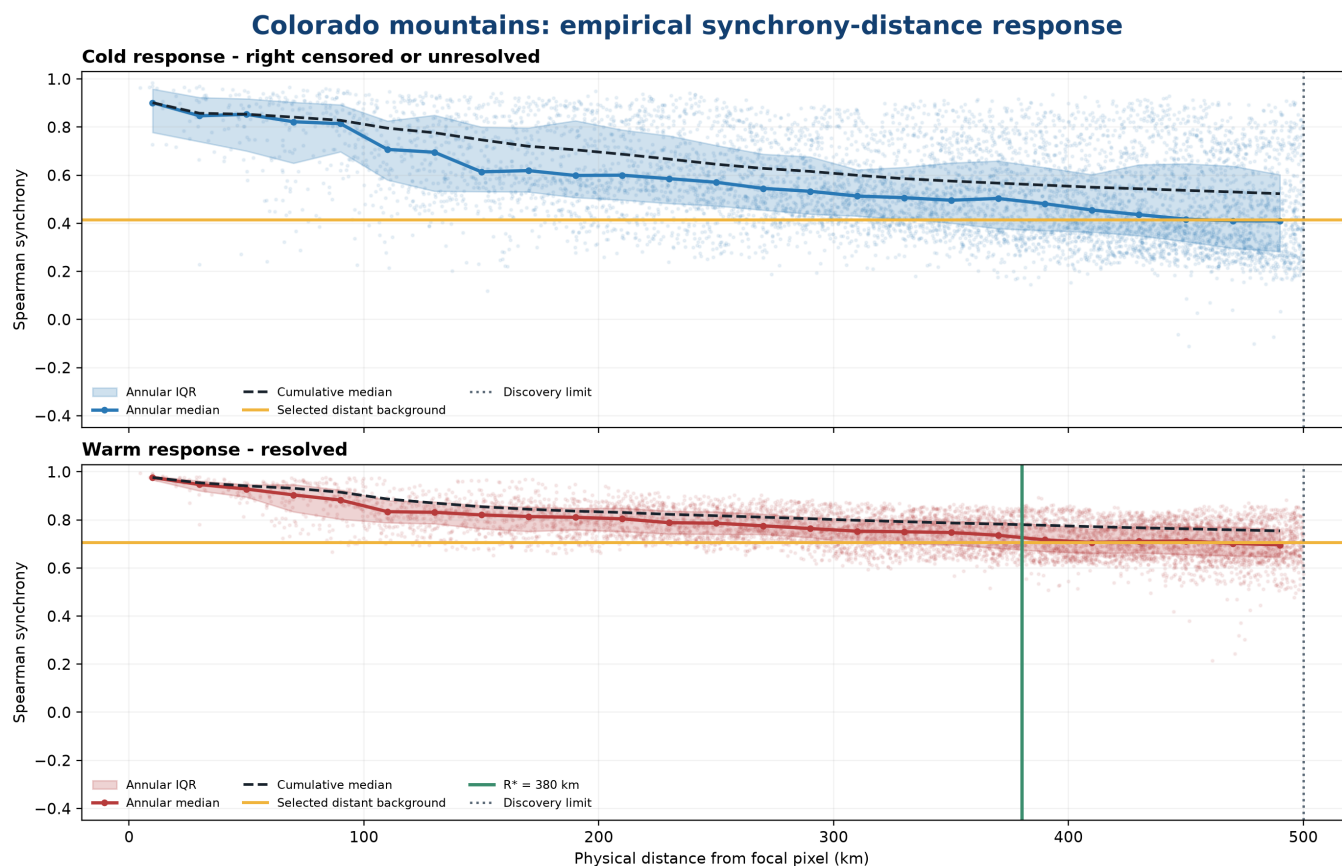
One canonical 500 km pair table supplies the fixed 100 km control, annular summaries, cumulative summaries, background comparisons, adaptive reductions, and every sensitivity test. Cold uses lower-tail TMIN; warm uses strictly upper-tail TMAX; $\Delta_S = S_{\text{cold}} - S_{\text{warm}}$.



Real synchrony-distance responses

6. REPRESENTATIVE EVIDENCE

Individual relationships remain visible as light points. The colored line is the annular median, the ribbon is its IQR, and the dashed line is the cumulative median. The curves are not forced to be monotonic or fitted to a parametric decay kernel.

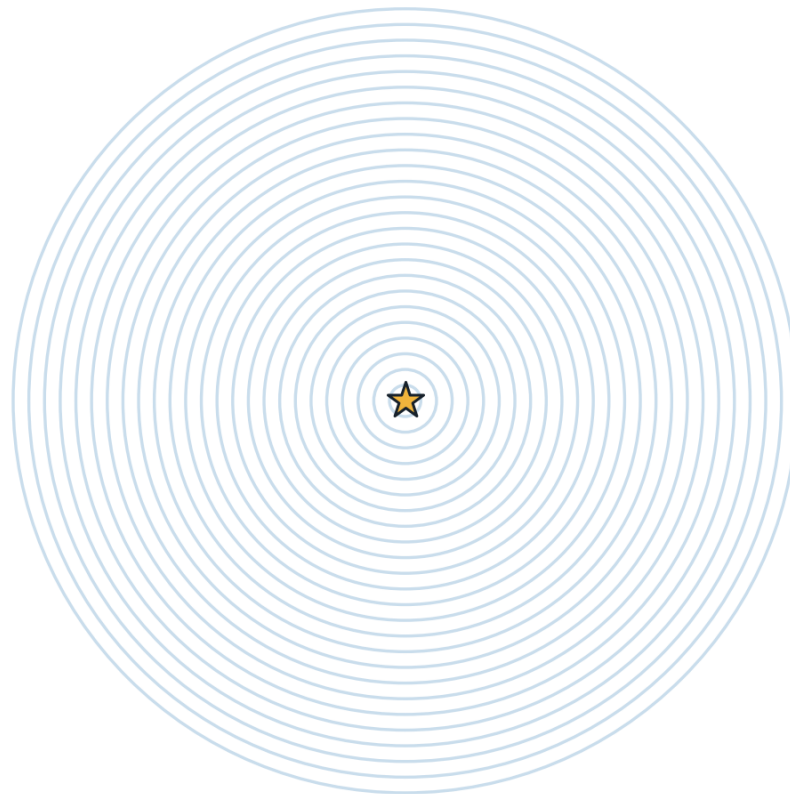


Summarize physical annuli

7. EMPIRICAL SUPPORT

The primary 20 km bins are wider than the roughly 4 km PRISM cells and contain ample relationships in the pilot. Counts are retained for every annulus. Unsupported bins cannot establish convergence.

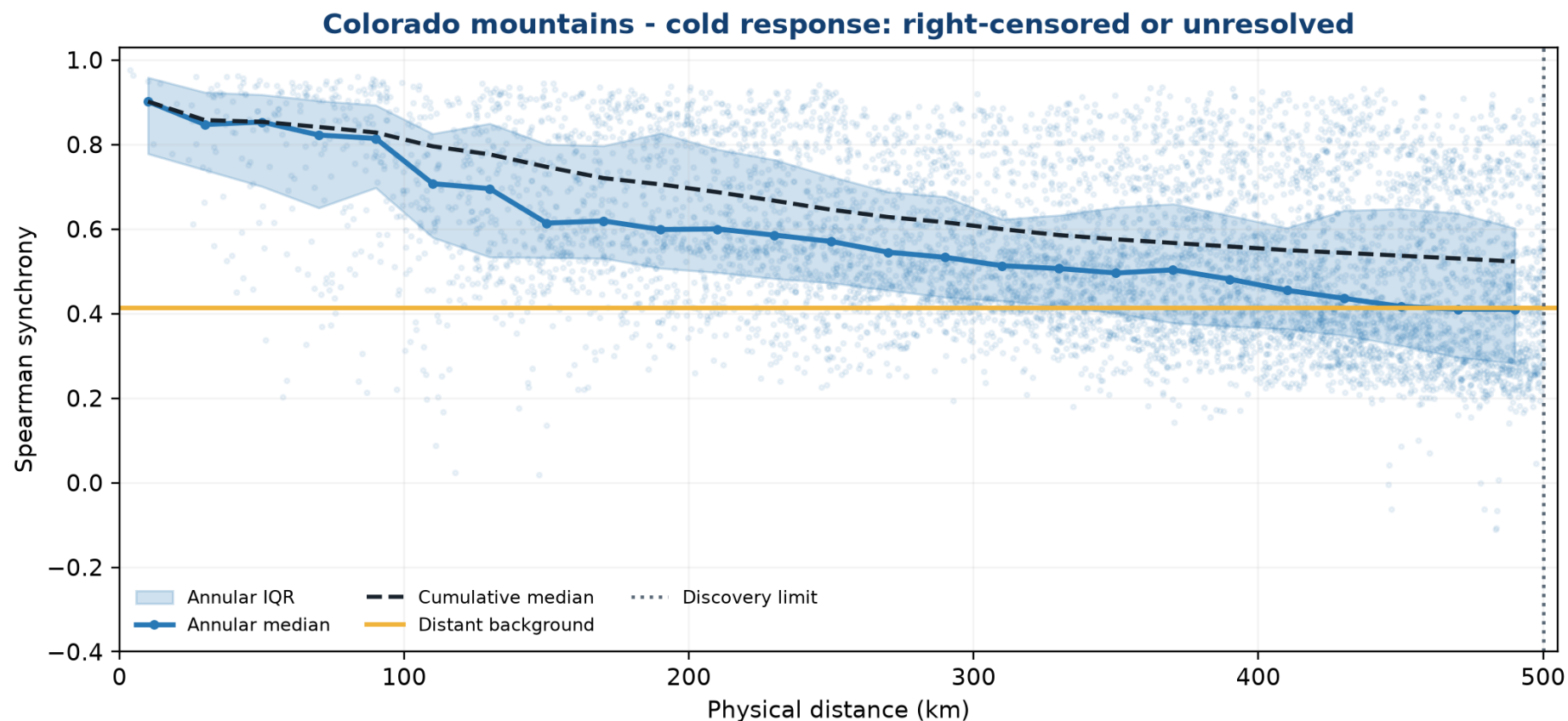
Twenty-kilometer physical annuli



Annular and cumulative views answer different questions

8. TWO CONVERGENCE TESTS

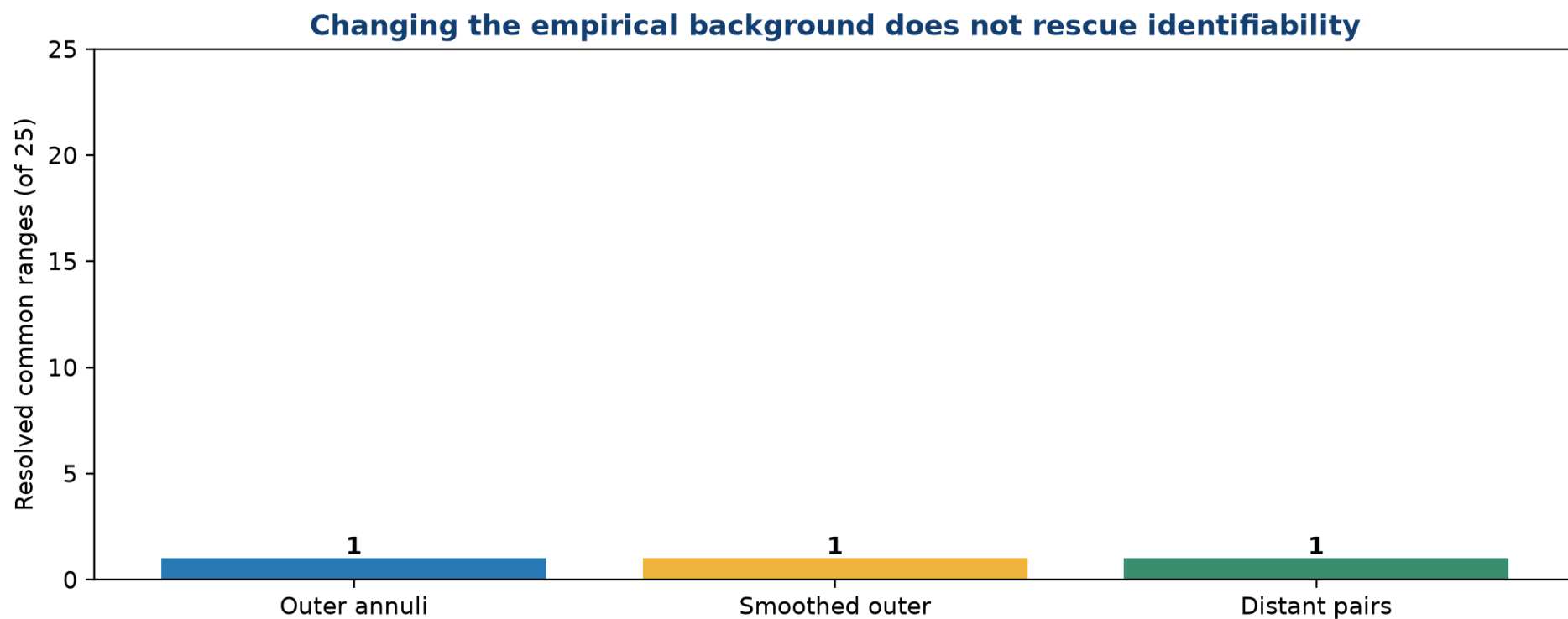
Annular medians ask whether newly encountered neighbors still differ from the distant background. Cumulative medians ask whether extending the neighborhood still changes the final collapse. Both must stabilize for three consecutive supported annuli.



Estimate a nonzero empirical background

9. BACKGROUND

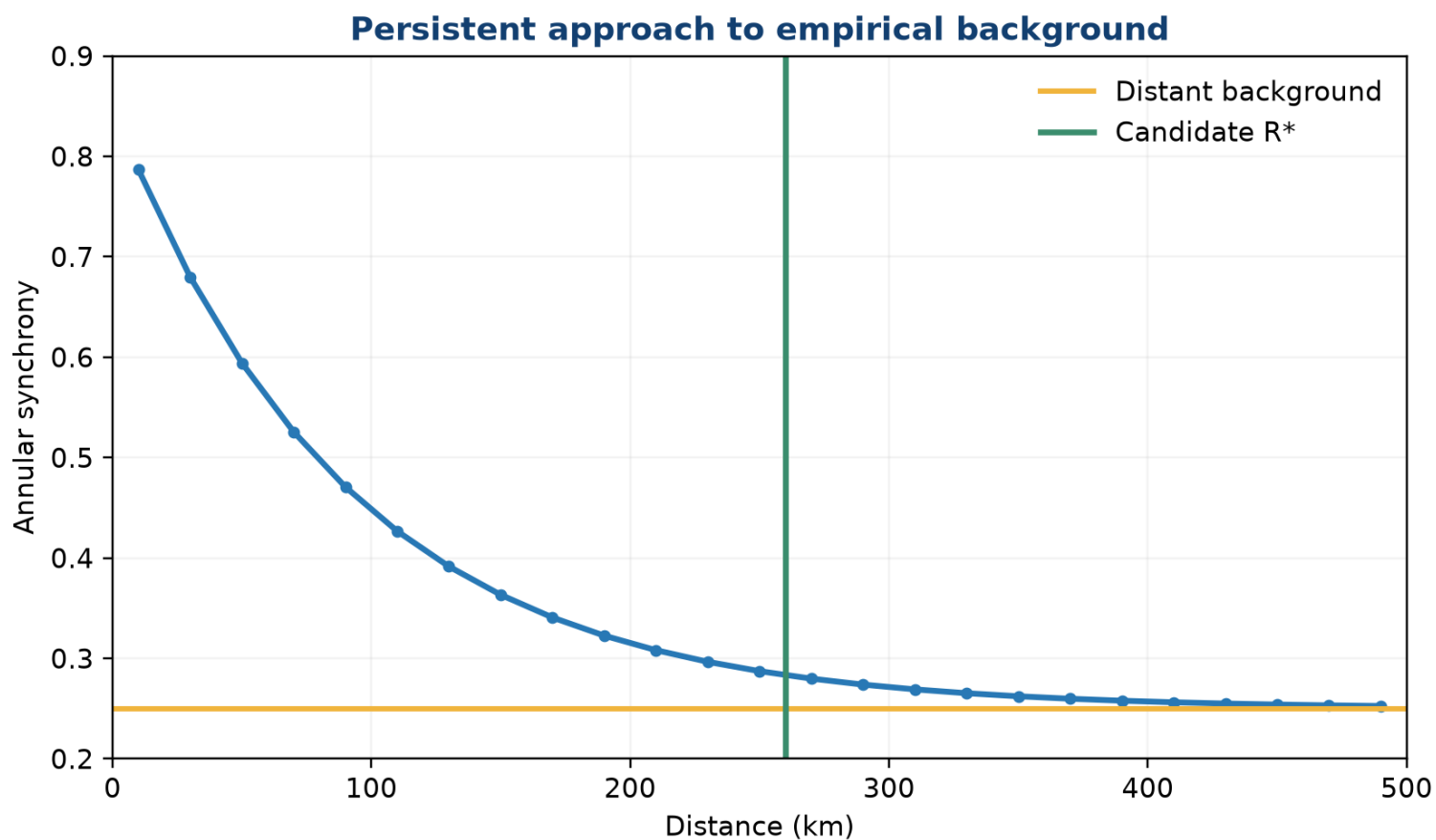
Broad-scale climate forcing can leave substantial synchrony at long distance. The method therefore estimates background rather than assuming zero. Outer annuli, smoothed outer annuli, and distant pairs are all retained and compared.



Identify R^* with persistence

10. DETERMINISTIC CRITERION

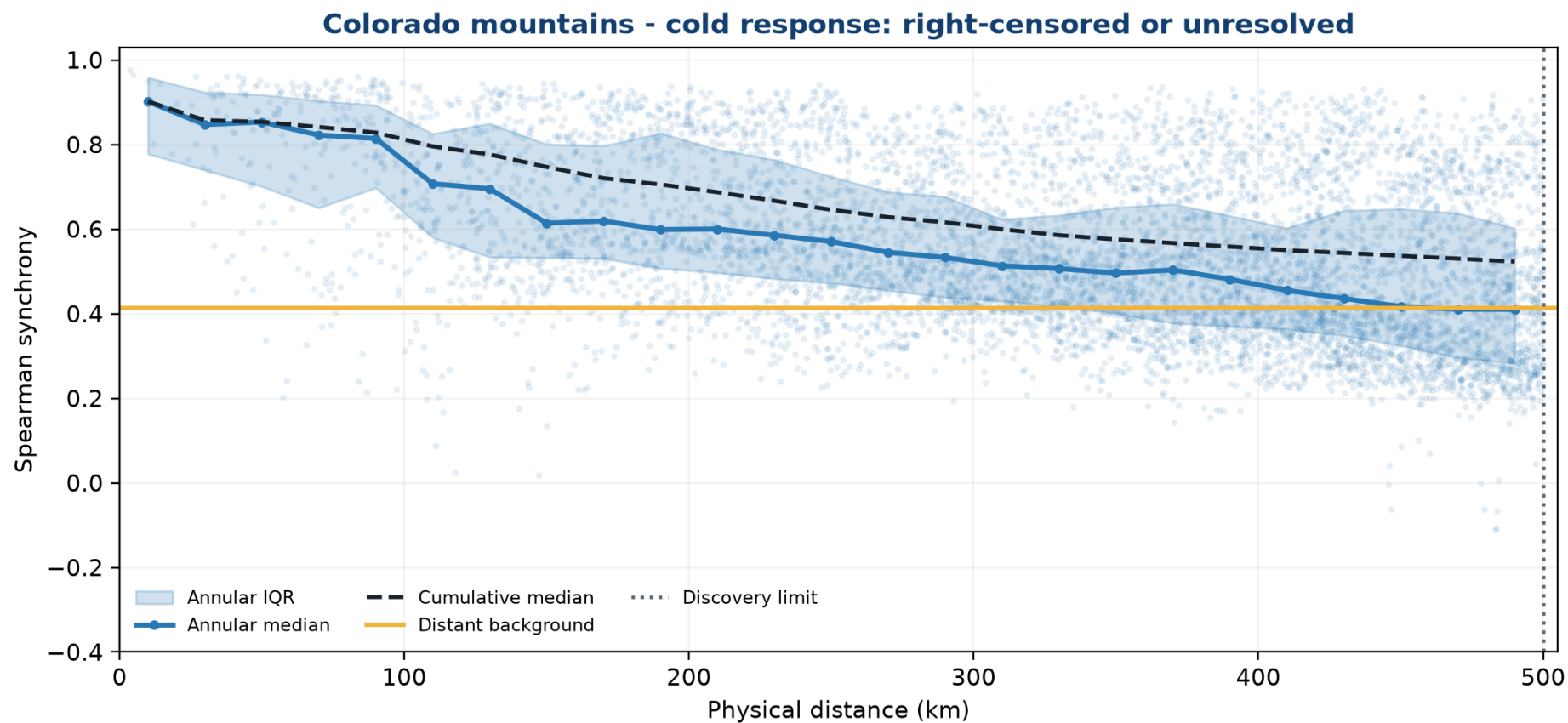
R^* is the first supported annulus whose median is near background and whose cumulative median is stable for three consecutive bins, with a later-shell safeguard. Candidates at or beyond 80% of discovery support remain unresolved.



Estimate cold range separately

11. R*COLD

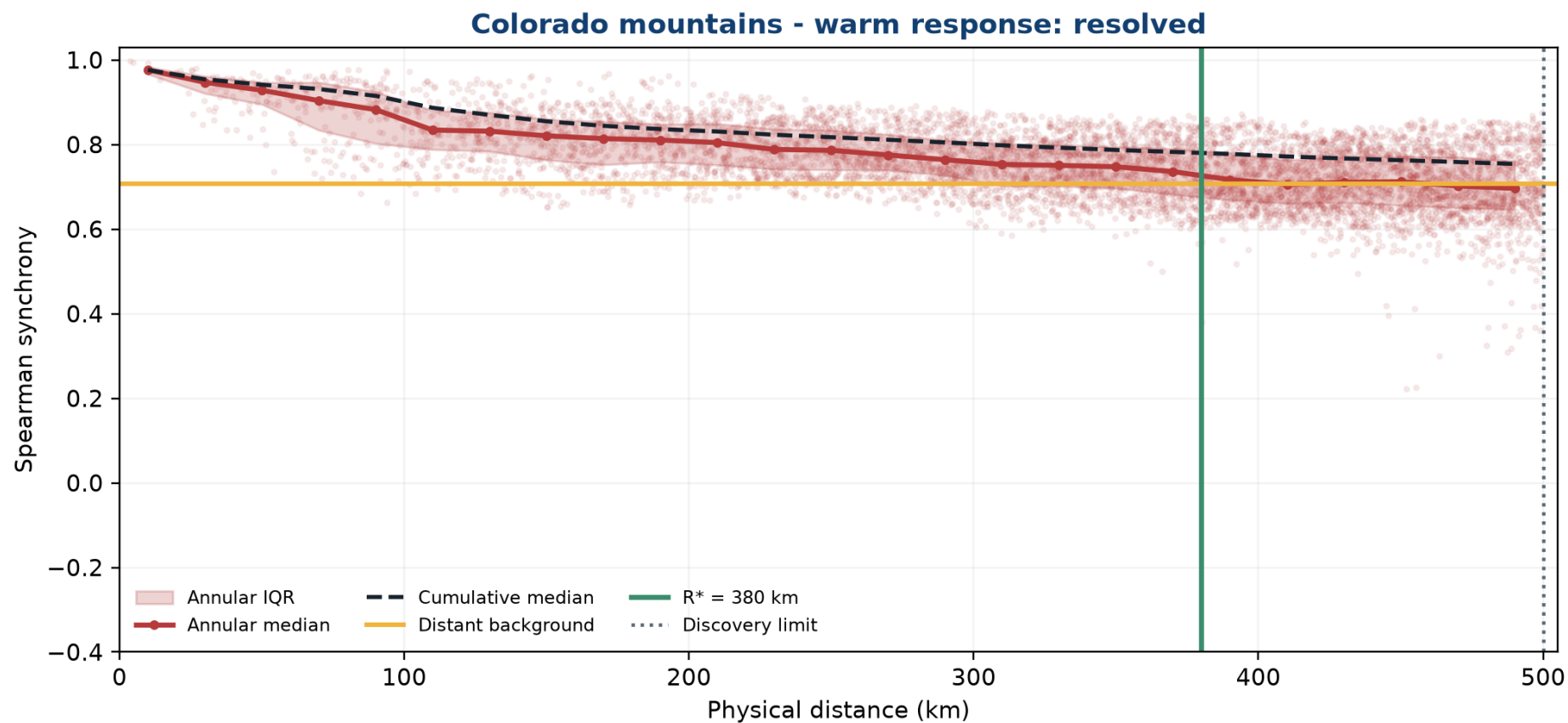
Cold range is inferred from joint lower-tail TMIN synchrony. In the mountain example the profile continues reorganizing with distance and is right-censored or unresolved at the primary setting.



Estimate warm range separately

12. R*WARM

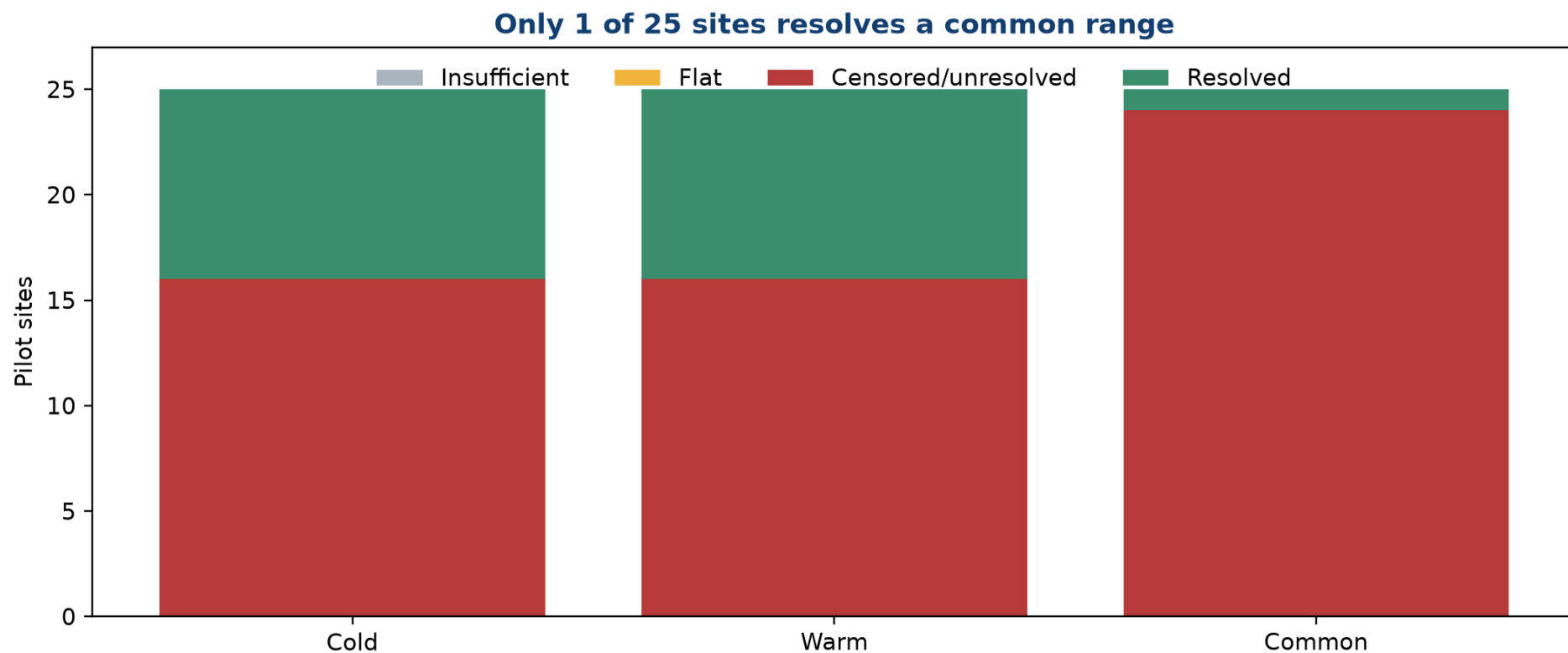
Warm range is inferred independently from joint upper-tail TMAX synchrony. The mountain warm response resolves at 380 km for this setting, but a tail-specific result cannot by itself support primary adaptive Delta.



Require a common neighbor set for Delta

13. R*COMMON

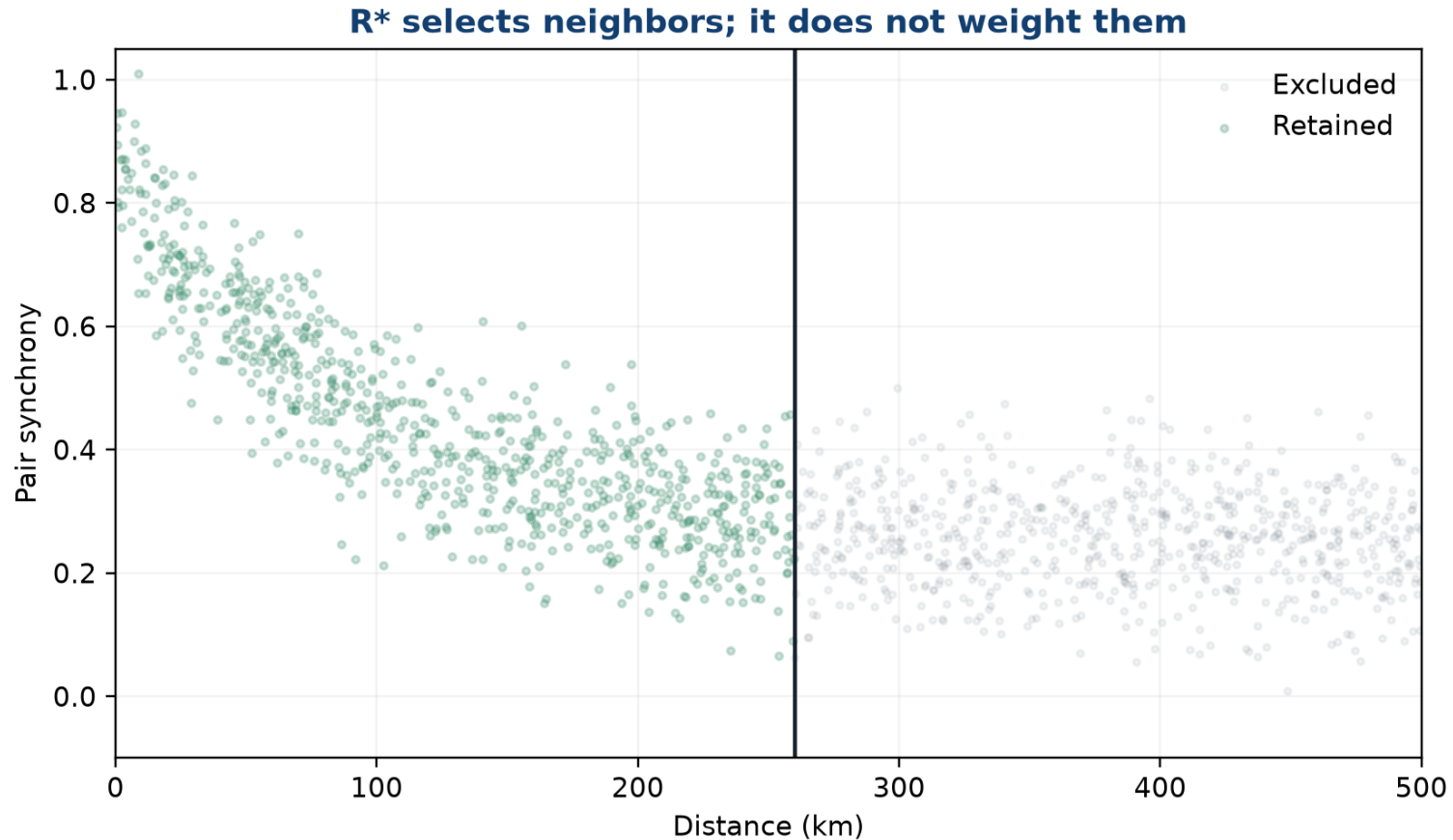
Primary adaptive Delta is defined only when both tail ranges resolve. Then $R^*_{\text{common}} = \max(R^*_{\text{cold}}, R^*_{\text{warm}})$, and cold, warm, and pairwise Delta are evaluated over exactly the same neighbors. Otherwise common range and adaptive Delta remain missing.



Use range for selection, not weighting

14. ADAPTIVE NEIGHBORHOOD

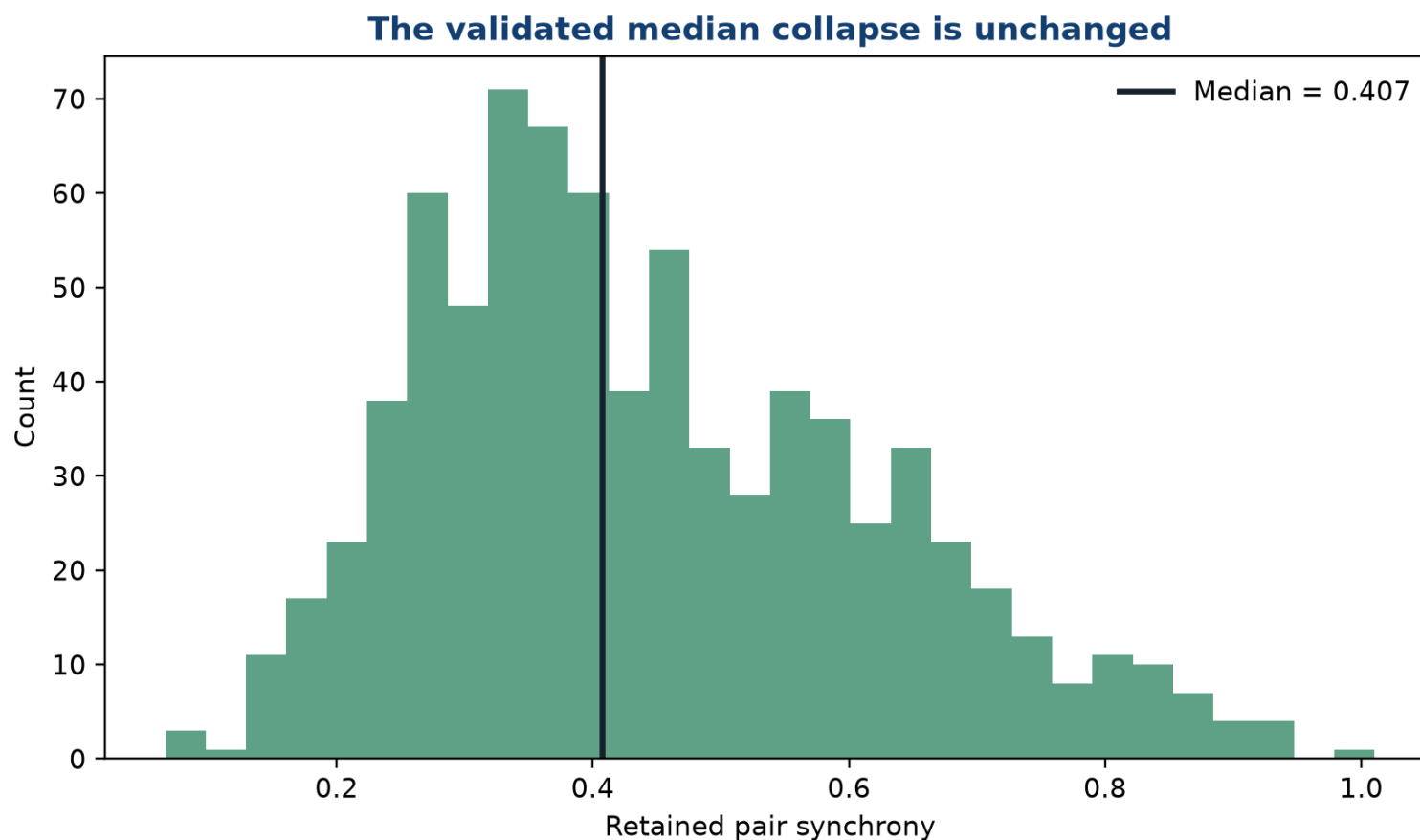
Pairs inside the resolved range are retained without distance weights; pairs outside are omitted. No exponential, Gaussian, power-law, or other kernel is imposed. The experiment isolates one change: neighborhood extent.



Apply the same robust collapse

15. ONE ADAPTIVE PIXEL

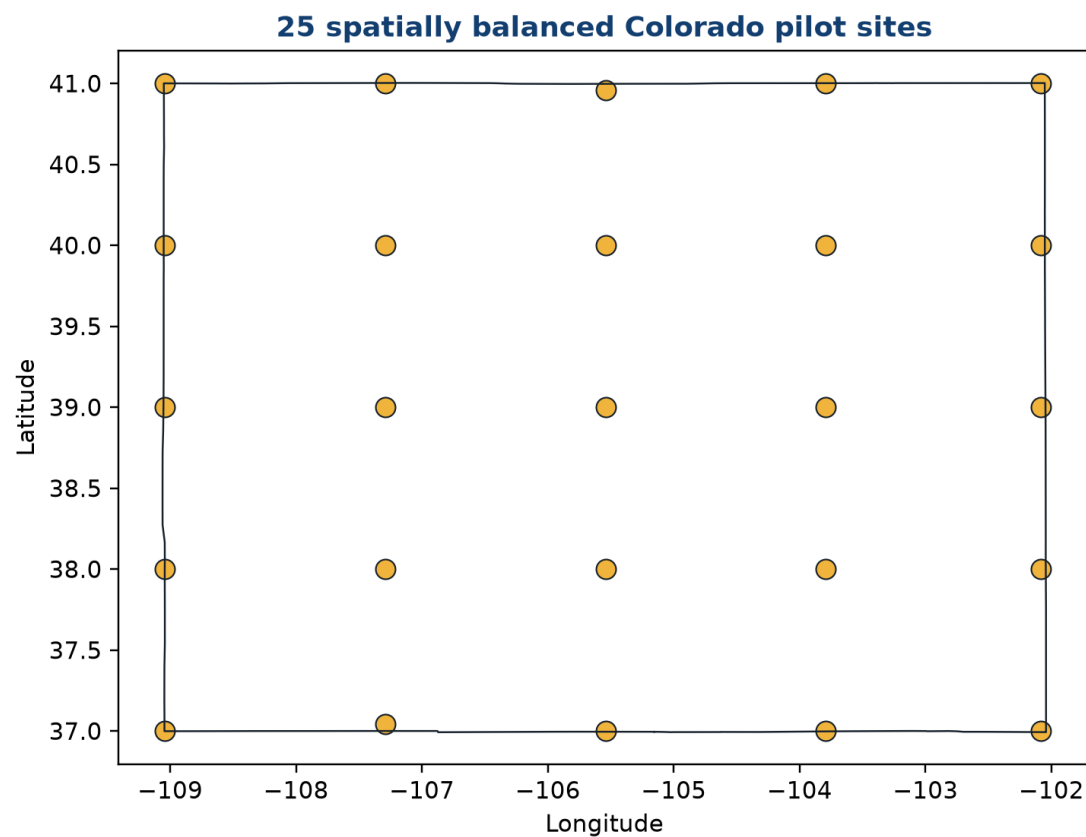
The retained relationships are reduced with the existing exact median. Primary adaptive Delta is $\text{median}[S_{\text{cold}}(i,j) - S_{\text{warm}}(i,j)]$ within R^* common; it is not the difference between two separately ranged marginal medians.



Repeat on a bounded Colorado pilot

16. SPATIAL GATE

Twenty-five spatially balanced focal cells cover Colorado plains, foothills, mountains, and high elevation. They are a method gate, not a statewide raster, and use the full CONUS snapshot for complete 500 km land support.

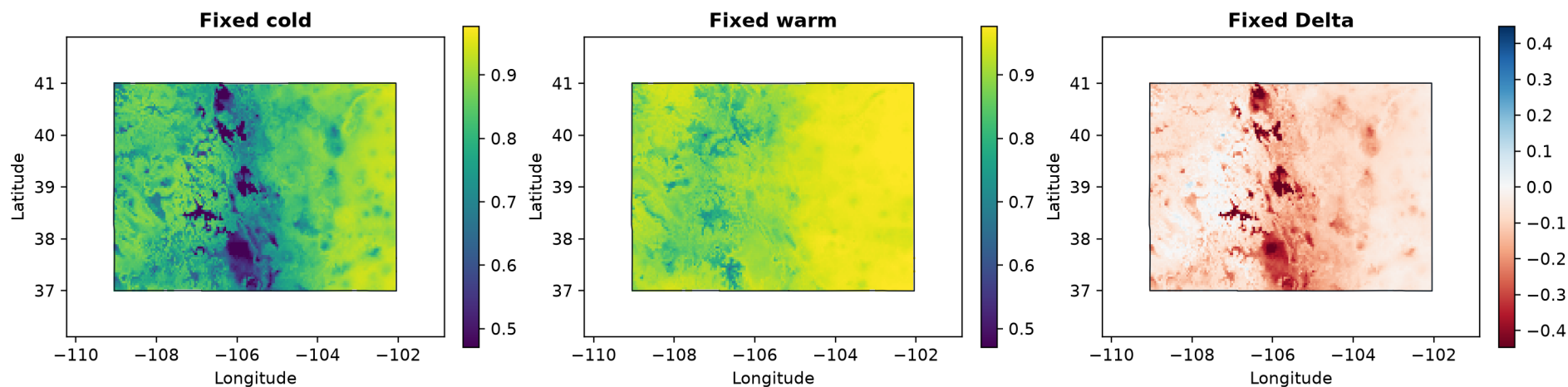


The fixed control remains available everywhere

17. FIXED PRODUCTS

The validated statewide 100 km cold, warm, and Delta maps are unchanged. The fixed Delta uses red for negative/warm-stronger and blue for positive/cold-stronger.

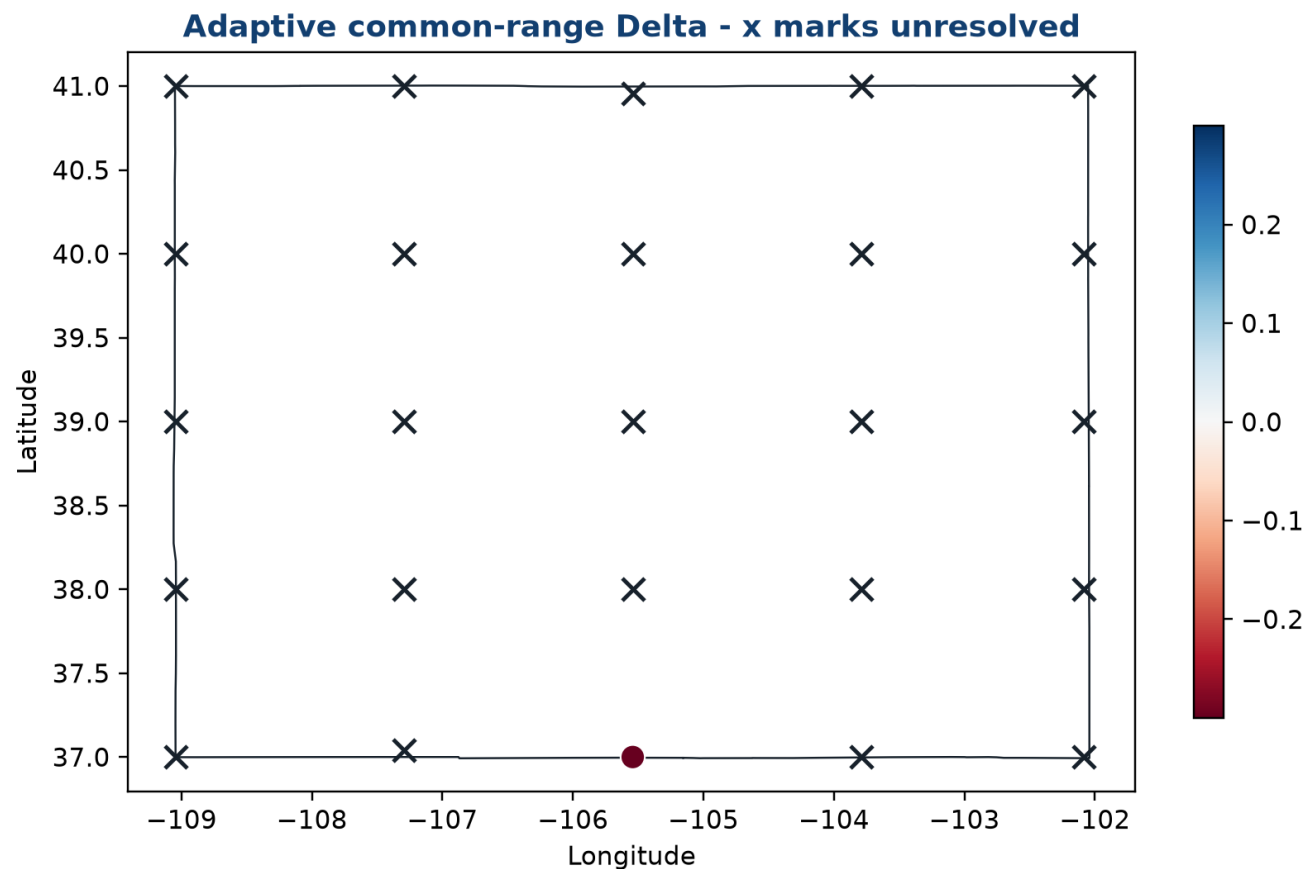
Validated full-state 100 km control remains unchanged



Adaptive Delta exists only where common range resolves

18. ADAPTIVE PRODUCT

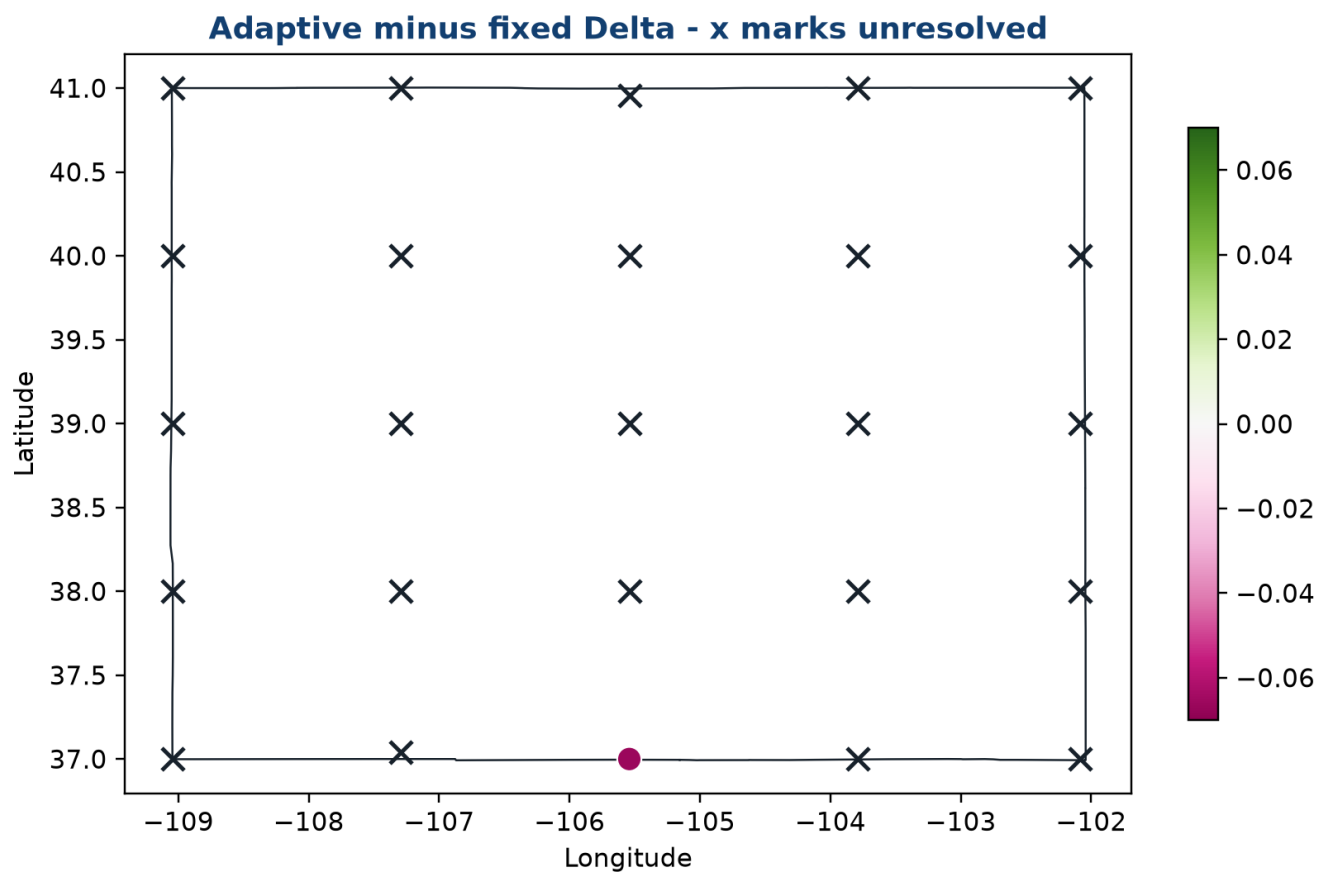
Only one of 25 pilot sites resolves both cold and warm ranges at the primary setting. Crosses mark unresolved sites. Filling those cells with 500 km would convert an observation boundary into a false scientific estimate.



Adaptive minus fixed Delta

19. DIFFERENCE PRODUCT

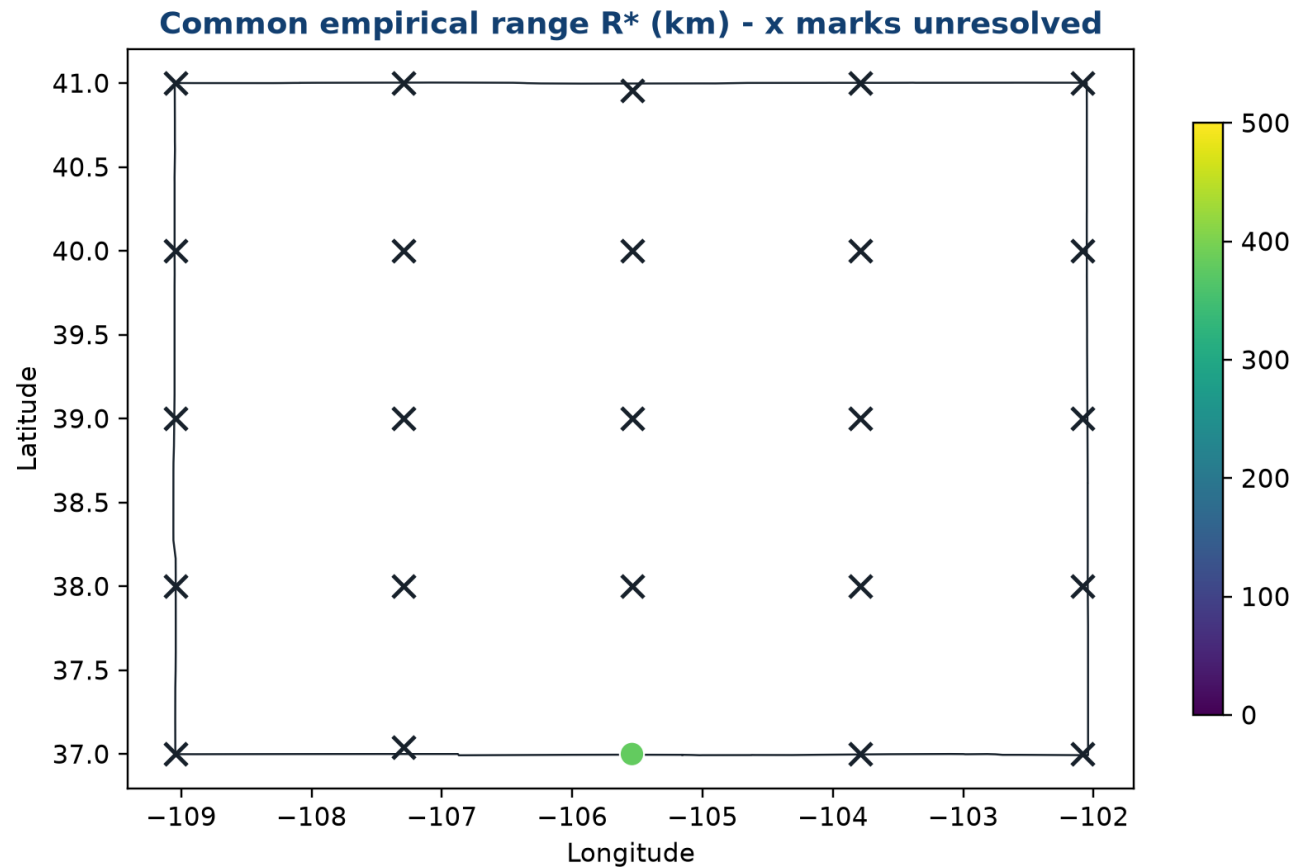
With one resolved common range, Pearson, Spearman, RMSE, sign-change frequency, and threshold exceedance rates are not scientifically estimable. The requested comparison statistics are retained with an explicit insufficient-support status.



Common synchrony-range map

20. PRIMARY RANGE PRODUCT

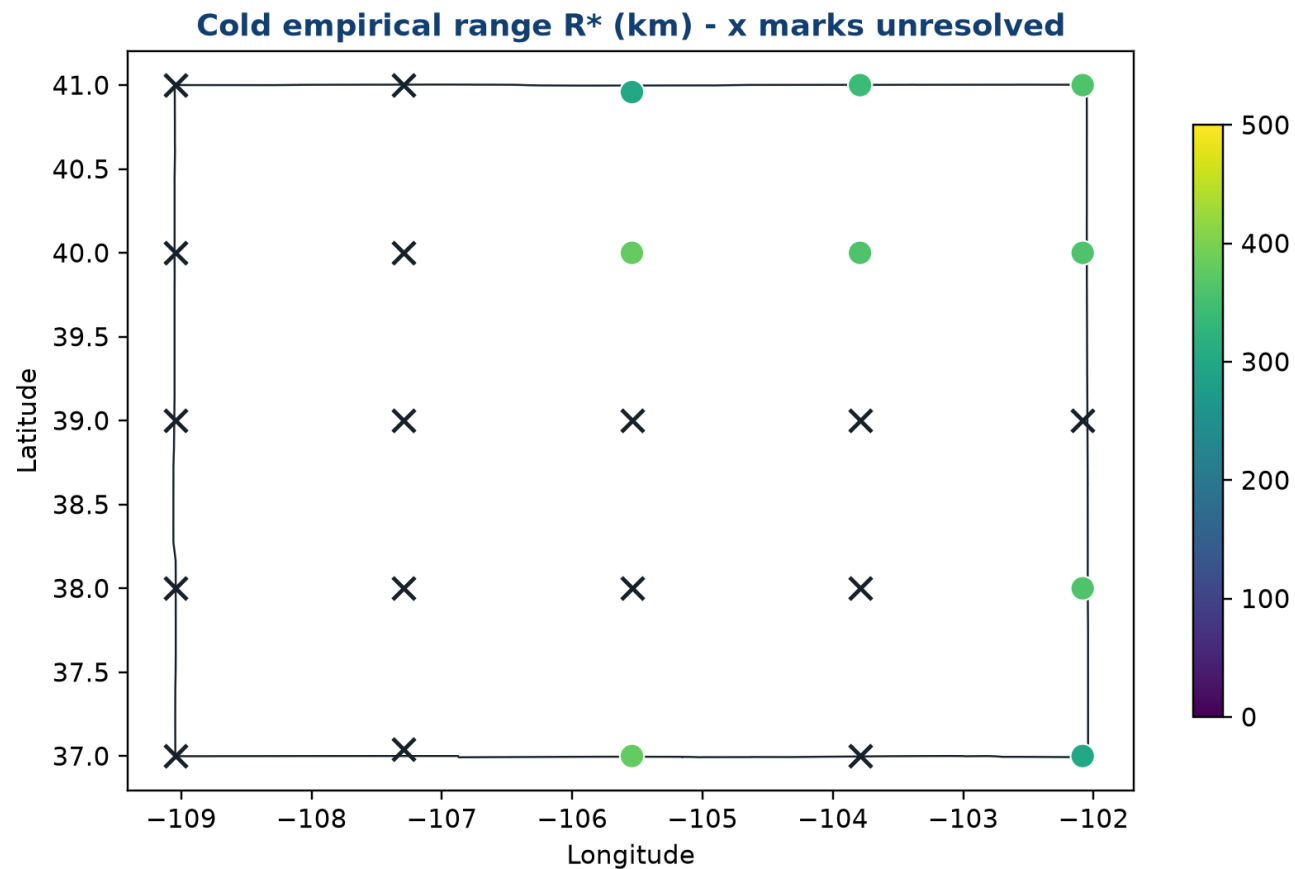
R^* common is shown only for the single site where both tail ranges resolve. The other 24 sites are censored or unresolved. This is the core reason a statewide product was not run.



Cold range map

21. TAIL-SPECIFIC RANGE

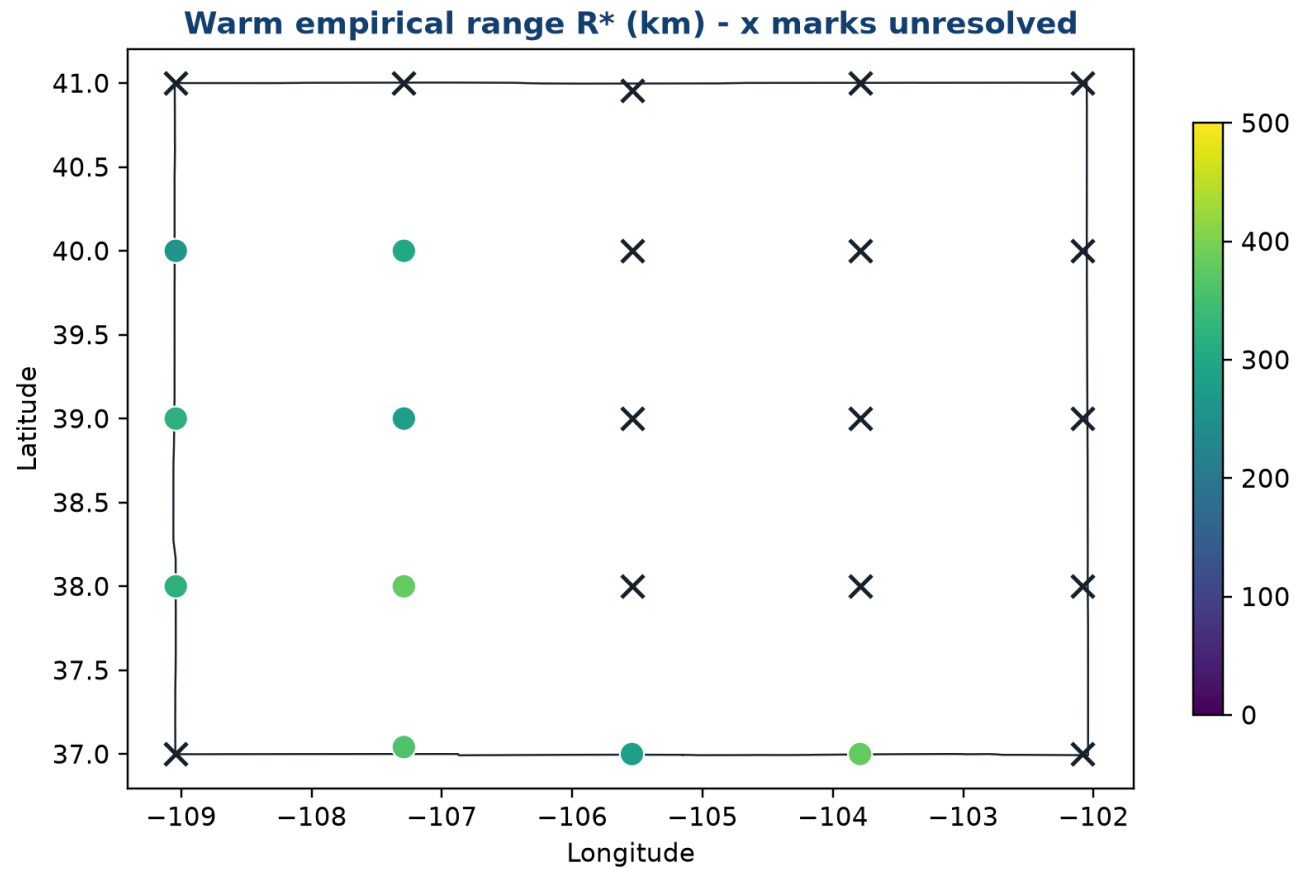
Nine of 25 pilot sites resolve a cold range, while the remainder are right-censored or unresolved. Tail-specific values are retained even when common range cannot be formed.



Warm range map

22. TAIL-SPECIFIC RANGE

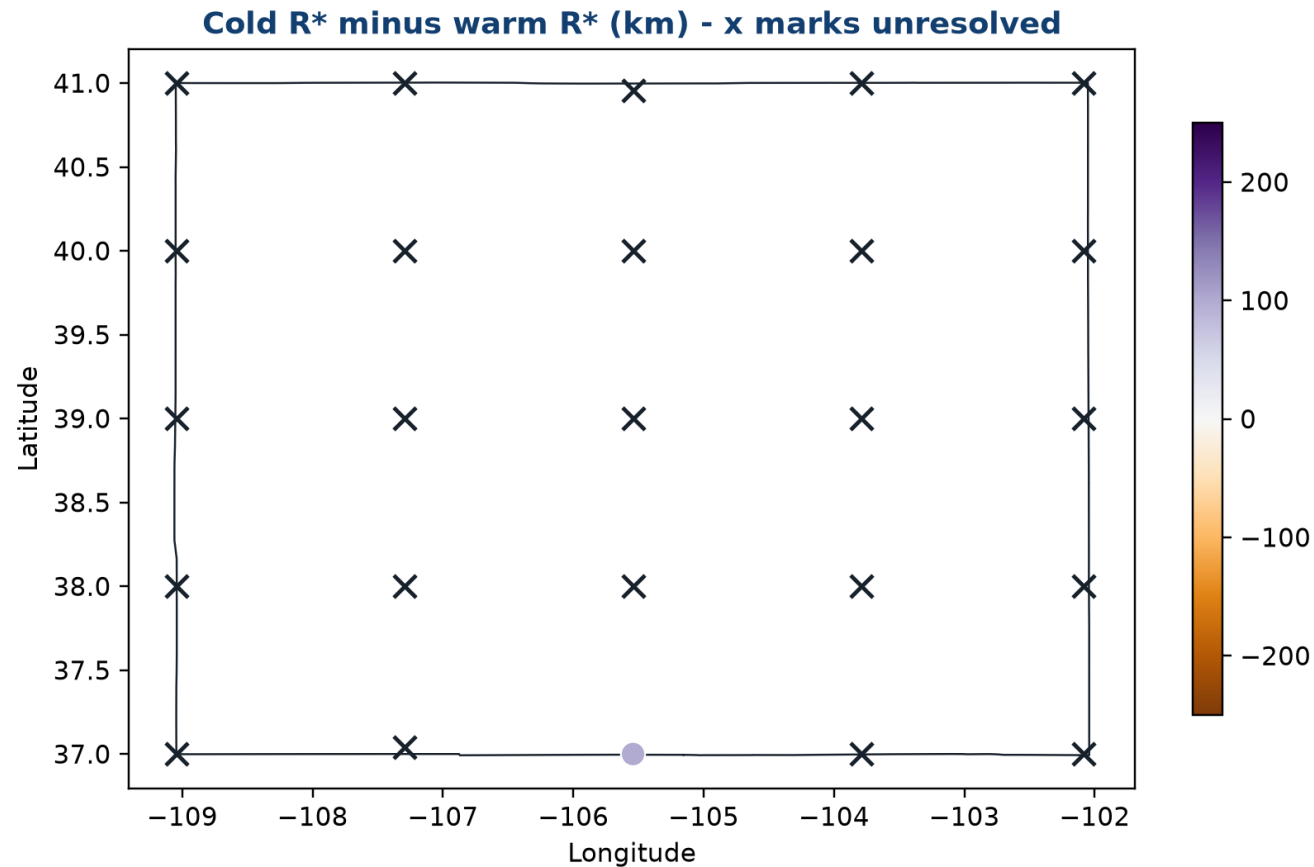
Nine of 25 pilot sites resolve a warm range, but often at different locations from cold. This supports keeping cold and warm range as separate scientific dimensions.



Cold-minus-warm range

23. DELTA_R IS NOT DELTA_S

$\Delta R = R_{\text{cold}} - R_{\text{warm}}$ can be calculated only where both ranges resolve. It measures spatial extent, not synchrony strength. The pilot provides too little common support for geographic interpretation.

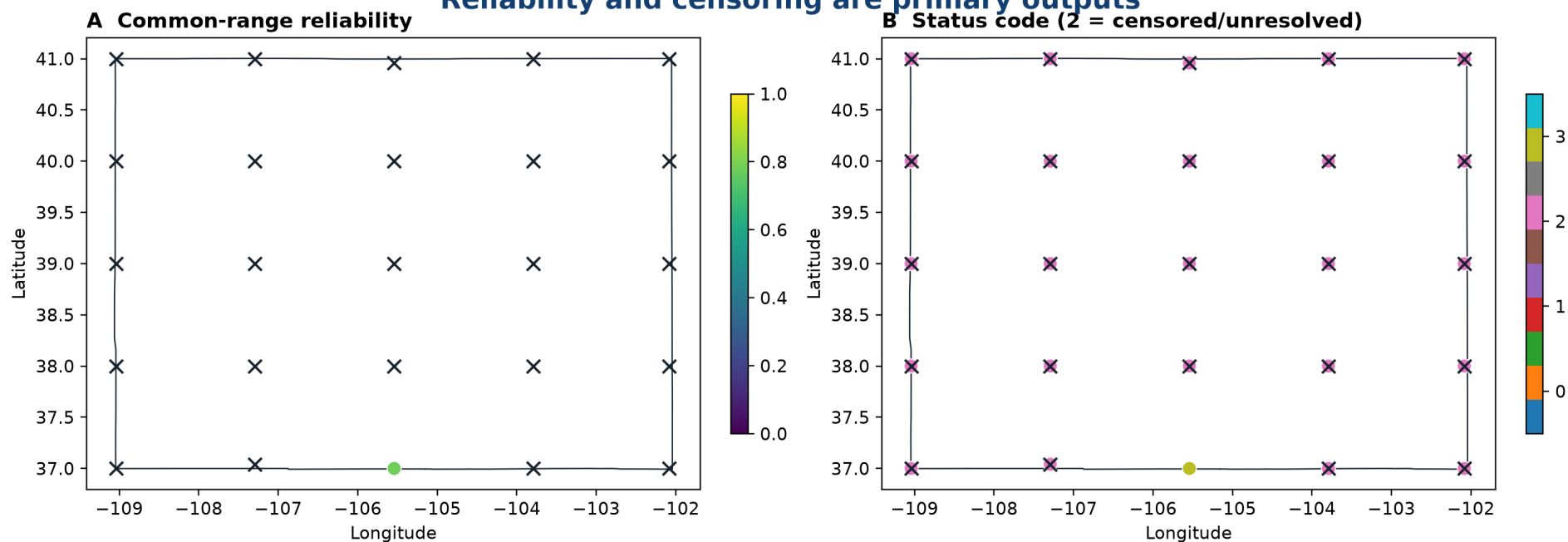


Uncertainty is part of the result

24. RELIABILITY AND CENSORING

The output retains a status code, boundary-censor flag, later-shell agreement, and reliability score. Reliability never overrides an unresolved status. Sector diagnostics also fail to establish scalar isotropy.

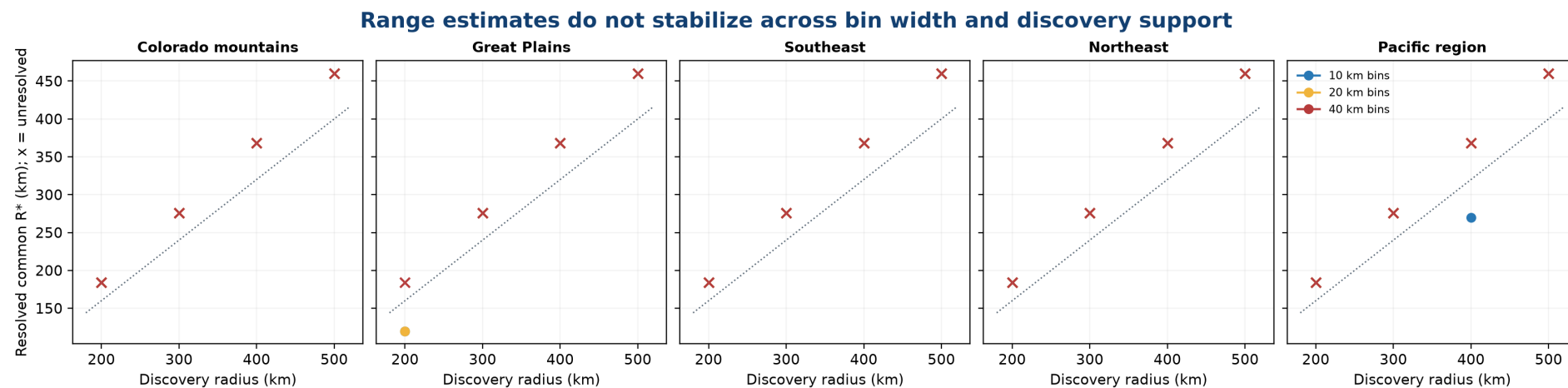
Reliability and censoring are primary outputs



Discovery-radius and bin-width sensitivity

25. STABILITY TEST

Ten, 20, and 40 km bins were tested under 200, 300, 400, and 500 km discovery limits using the same 500 km pairs. Crosses denote unresolved results. No Colorado pilot site stabilized between 400 and 500 km at the primary bin width.

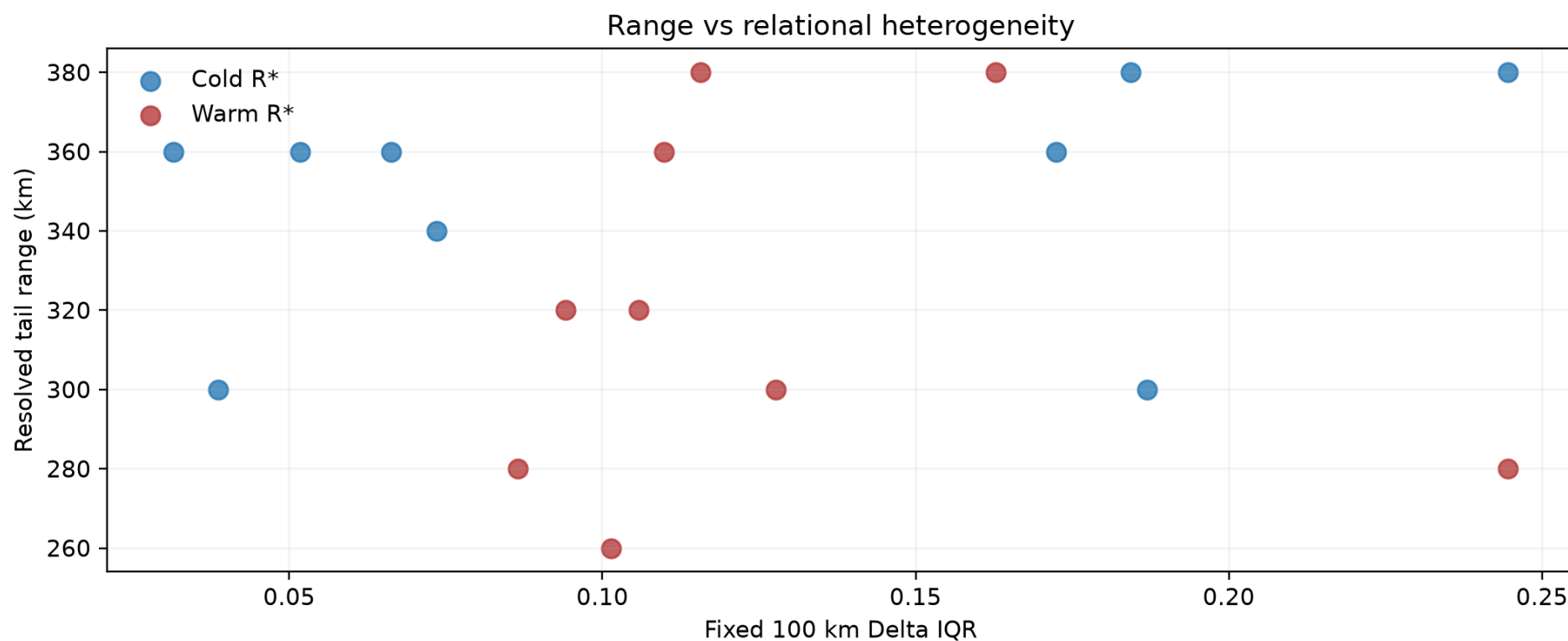


Relationship with Delta IQR

26. EXISTING QC

Tail-specific resolved ranges can be plotted against fixed-radius relational heterogeneity, but common-range coverage is too sparse for a valid map-level association. The pilot therefore makes no claim about whether short or long range tracks high Delta IQR.

Common-range inference is too sparse for a valid map-level relationship

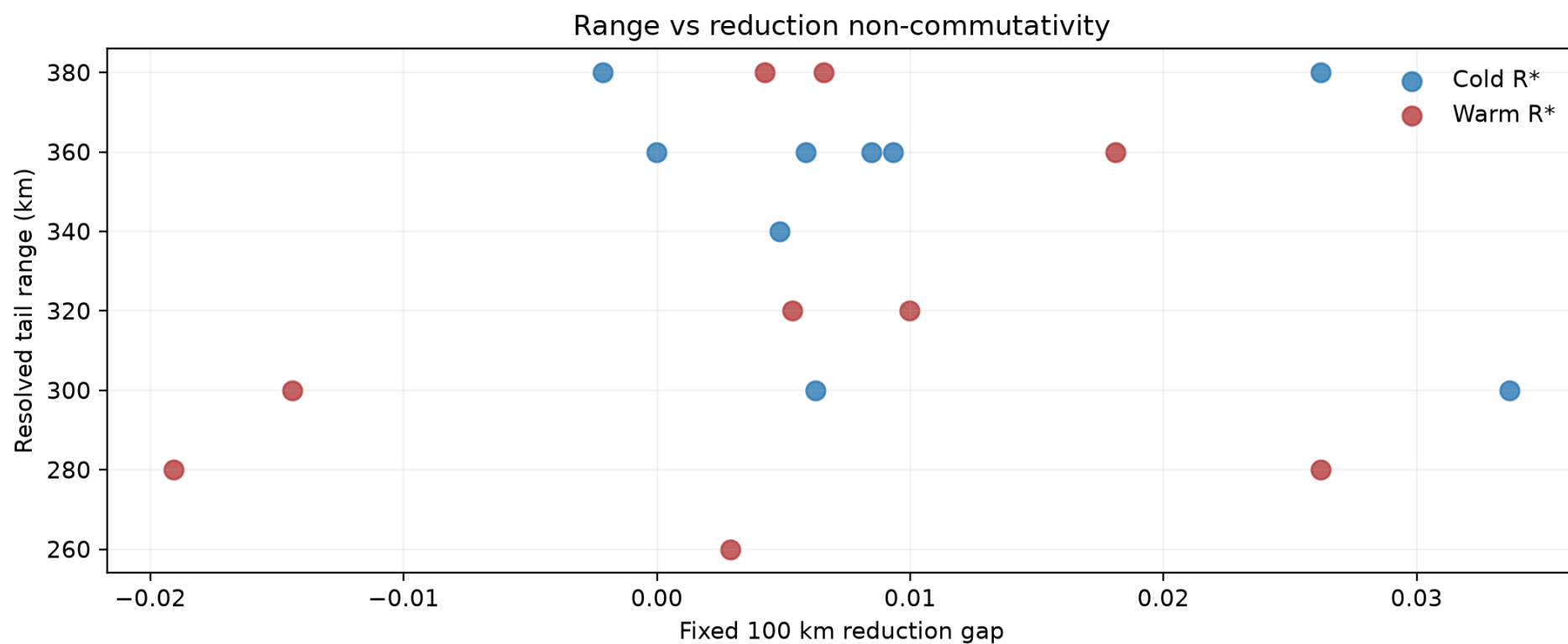


Relationship with the reduction gap

27. EXISTING QC

The reduction gap measures non-commutativity between median pairwise Delta and the difference of marginal medians. It remains conceptually separate from empirical range, and the current pilot cannot establish a robust relationship.

Common-range inference is too sparse for a valid map-level relationship



What changed - and what did not

28. CONTRACT BOUNDARY

The new work adds empirical distance-response summaries and a guarded neighbor-selection rule. It does not revise historical fixed-radius values, change cold/warm tail semantics, add a kernel weight, or reinterpret range as dispersal.

Changed	Unchanged
Large discovery support	Validated PRISM observations and pair kernel
Annular plus cumulative summaries	Cold $\leq$ TMIN median; warm $>$ TMAX median
Cold/warm/common R* statuses	$\Delta_S = S_{\text{cold}} - S_{\text{warm}}$
Adaptive neighbor inclusion	Exact median pairwise-Delta collapse
Reliability and censoring	No stacking, second convolution, or distance weights

Decision: HOLD before full Colorado or CONUS

30. INTERPRETATION AND NEXT STEP

The empirical curves are real and geographically structured, but the predeclared scalar range criterion is not identifiable or stable often enough to support adaptive production maps. The 25-site pilot evaluated 1.18 million nonself pairs; planning bounds reach 0.38-0.77 billion for Colorado and 11.35-22.69 billion for CONUS. No full Colorado or CONUS adaptive-range production run was performed. Reproduce with:

```
.venv/bin/python scripts/run_empirical_synchrony_range_pilot.py
```

Next: test a revised, predeclared empirical criterion or longer independent windows. Do not tune thresholds merely to increase map coverage.

Gate	Observed
Common resolved >= 70%	4% (1 of 25)
Discovery-stable >= 70%	0% (0 of 25)
Representative resolved >= 4 of 5	0 of 5
Scale-up decision	HOLD